

VL 300: Reverse Engineering - E-Waste Management

Course Code: VL 300

Course Name: Reverse Engineering - E-Waste Management

L-T-P-C: 0.5-0-1-1

Prerequisites:

Students intended for: B.Tech

Elective or Core: Core

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Introduction to Reverse Engineering and e-Waste:** Definition and Importance of Reverse Engineering, Basics of E-Waste: Definition, Sources, and Types, Global and Indian e-Waste Statistics, Environmental and Health Impacts of e-Waste, Regulatory Frameworks: Basel Convention, e-Waste Management Rules (India)
- **e-Waste Generation and Composition Analysis:** Lifecycle of Electronic Devices, Major Contributors to E-Waste (Smartphones, Laptops, PCBs, Batteries, etc.), Material Composition of E-Waste (Metals, Plastics, Toxic Elements), Techniques for Identifying Valuable and Hazardous Components
- **Fundamentals of Reverse Engineering in E-Waste Recycling:** Concept and Applications of Reverse Engineering, Tools and Software for Reverse Engineering (CAD, Simulation Tools), Disassembly Techniques for Electronic Components, Identification and Reuse of Functional Parts, Case Study: Reverse Engineering a Mobile Phone or a Laptop
- **Material Recovery from E-Waste:** Mechanical Processing (Shredding, Crushing, Sorting), Thermal Processing (Pyrolysis, Incineration), Hydrometallurgical and Bio metallurgical Methods, Precious Metal Recovery (Gold, Silver, Copper), Challenges in Efficient Material Recovery
- **Sustainable E-Waste Management Strategies:** Circular Economy and Sustainable Product Design, Eco-friendly Recycling Technologies, Extended Producer Responsibility (EPR), Urban Mining and Resource Efficiency, Case Study: Successful E-Waste Recycling Companies
- **Practical Aspects of Reverse Engineering for E-Waste:** Hands-on Demonstration of Disassembly and Component Analysis, Identification of Reusable and Recyclable Components, PCB Reverse Engineering Techniques, DIY Repair and Refurbishment of Old Electronics, Lab Activity: Disassemble and Analyse an E-Waste Item
- **Future Trends and Innovations in E-Waste Management:** Emerging Technologies for E-Waste Processing, Role of AI, IoT, and Blockchain in E-Waste Management, Policy and Business Opportunities in E-Waste Recycling, Career Opportunities in E-Waste Management, Future of E-Waste Management and Reverse Engineering

Laboratory/practical/tutorial Modules:

Reverse Engineering Reports on Disassembled Devices, Real-World E-Waste Management Models, Hands-on Experience in Reverse Engineering & Recycling, Innovative E-Waste Management Solutions

Text books:

1. Vannessa Goodship, Ab Stevels, **Waste Electrical and Electronic Equipment (WEEE) Handbook**, Woodhead Publishing, 2012.
2. Rakesh Johri, **Electronic Waste Management**, TERI Press, 2008.
3. Wego Wang, **Reverse Engineering: Technology of Reinvention**, CRC Press, 2010.
4. Klaus Hieronymi, Ramzy Kahhat, Eric Williams, **E-Waste Recycling and Management: Present Scenarios and Future Perspectives**, Springer, 2013.

References:

1. Kuan Yew Cheong, Mohammad Tariqur Rahman, **Sustainable Materials and Green Processing for Energy Conversion**, Elsevier, 2021.
2. Varsha Bhagat-Ganguly, **Electronic Waste Management: Issues and Strategies**, Routledge, 2021.
3. Ernst Worrell, Markus A. Reuter, **Handbook of Recycling: State-of-the-Art for Practitioners, Analysts, and Scientists**, Elsevier, 2014.
4. Bhuvan Unhelkar, **Green IT Strategies and Applications: Using Environmental Intelligence**, CRC Press, 2011.